

$$r_{pb} = \frac{\mu_{nor} - \mu_{un}}{\sigma_{total}} \sqrt{\frac{N_{nor} N_{un}}{(N_{nor} + N_{un})^2}}$$

$$t = r_{pb} \sqrt{\frac{(N_{nor} + N_{un}) - 2}{1 - r_{pb}^2}}, \quad df = (N_{nor} + N_{un}) - 2$$

$$p_{\text{two-sided}} = 2 \left[1 - F_{t_{df}}(|t|) \right]$$